

- Q.27 Describe the methods of providing the "Reinforcement in Footings" as per IS code: 13920-1993, to act as an earthquake resistant structural member.
- Q.28 Describe the various grades of steel recommended by IS: 13920-1993 to be provided for construction of building to make them as earthquake resistant buildings.
- Q.29 Describe the general principles to be followed during the construction of a conventional brick masonry buildings to make them as earth quake resistance buildings.
- Q.30 Describe the purposes of providing the "Through stones" in stone masonry walls.
- Q.31 Describe the suitable technique to be used for strengthening the R.C.C. Columns, damaged by an earthquake.
- Q.32 Describe the procedure of repair of cracks developed due to an earthquake, in a residential masonry building.
- Q.33 Describe the various phases of "Rescue by Steps" for disaster management.
- Q.34 Describe the various classifications of casualties.
- Q.35 Enlist the various rescue equipments required to carry out the rescue operations, for damage control and casualty management in a hospital consist of five floors.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the construction and working of a Seismograph with the help of a diagram.
- Q.37 Explain the necessity of providing the seismic bands in a masonry building construction to make it an earthquake resistant building, with the help of diagram.
- Q.38 Explain the ways in which the "Irregularity in Building Plan or Shape" affects their seismic performance.

No. of Printed Pages : 4

Roll No.

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6th Sem / Civil, Brick Tech (Elective) Constr. Mgmt/ Civil Engg (Spl. Highway Engg.)

Sub : Earthquake resistant building Construction

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which type of fault is associated with horizontal movement of tectonic plates?
- a) Normal fault b) Reverse fault
c) Strike slip fault d) Thrust fault
- Q.2 Which of the following is the other term used for "Focus" in relation with an earthquake?
- a) Epicentre b) Hypocentre
c) Isocentre d) Principle Point
- Q.3 The fundamental mode of vibrating of a structure is the mode having the _____ natural frequency.
- a) Lowest b) Average
c) Highest d) None of the above
- Q.4 When there is a reduction in amplitude over every cycle of vibration, then the body is said to have _____.
- a) Free vibration b) Forced vibration
c) Damped vibration d) Undamped vibration
- Q.5 During an earthquake, soil and water can combine to form a semi-solid material in a process called _____
- a) Liquefaction b) Liquefying
c) Liquefaction d) Quick sanding
- Q.6 _____ is an essential attribute of an earthquake resistant design of a structure that serves as a shock absorber in a

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structure and reduces the transmitted to one that is sustainable.

- a) Elasticity b) Ductility
- c) Plasticity d) All of the above

Q.7 According to IS: 1893-2002 (Part-I) the zone factor for seismic zone-III is equal to ____

- a) 0.10 b) 0.16
- c) 0.24 d) 0.36

Q.8 ____ is the process of adding strength to any part of an existing building to provide higher strength than its original.

- a) Re-strengthening b) Repair
- c) Retrofitting d) Rehabilitation

Q.9 Who is the chairman of National Disaster Management Authority?

- a) President b) Prime Minister
- c) Defence Minister d) Home Minister

Q.10 In which city the National Institute of Disaster Management is situated?

- a) Kolkata b) New Delhi
- c) Hyderabad d) Manipur

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 More the flexibility of building ____ is the value of fundamental period "T" (Lower / Higher)

Q.12 The dissipation of energy generated in the structure by the effect of ground movement is called ____ (Seismic energy dissipation / Non-seismic energy dissipation)

Q.13 The building that lacks symmetry and has discontinuity in geometry, mass or load resisting elements is termed as ____ (Regular building / Irregular buildings)

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Q.14 As per IS 1893:2002 India is divided into ____ seismic zones. (Four/Five)

Q.15 ____ of steel due to environmental attack which can result in reduction of area of steel and further load carrying capacity. (Corrosion / Carbonation)

Q.16 Columns not having footing resting on ground, is called ____ (Pile foundation / Shallow foundation)

Q.17 Small size cracks (less than 5mm width) can be repaired by ____ technique. (Epoxy injection / Shotcreting)

Q.18 In ____ method, the structure and foundation are separated by bearing or flexible layer. (Base isolation / Shear wall)

Q.19 Mitigation aims to ____ the loss of human life that would result from a disaster. (Increase / Minimize)

Q.20 Strengthening and retrofitting are the aspects of disaster management that involve making structure ____ resilient to hazards. (Less/ More)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Describe the different types of seismic waves with the help of diagrams.

Q.22 Write the differences between earthquake magnitude and earthquake intensity

Q.23 Describe the "Connection Failure" as a mode of failure during an earthquake.

Q.24 Write a short note on "Seismic Zoning Map of India".

Q.25 Describe the causes of failure of non-structural components of traditionally built-up constructions in India; due to an earthquake.

Q.26 Enlist the various seismic codes published by Bureau of India Standard.

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